7 You are given that $\mathrm{P}(A) = 0.6$, $\mathrm{P}(B) = 0.5$ and $\mathrm{P}(A \cup B)' = 0.2$.

(a) Find $\mathrm{P}(A \cap B)$. **[2]**

(b) Find $\mathrm{P}(A \mid B)$. **[2]**

(c) State, with a reason, whether A and B are independent. **[1]**